



Physiology of the Liver

Physiology GIT Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

1. What are the two blood supplies to the liver, and what percentage does each contribute?
2. Name three things the liver does NOT do that students often mistakenly assume it does.
3. What's the difference between an ALT/AST pattern and an ALP/GGT pattern on liver function tests?

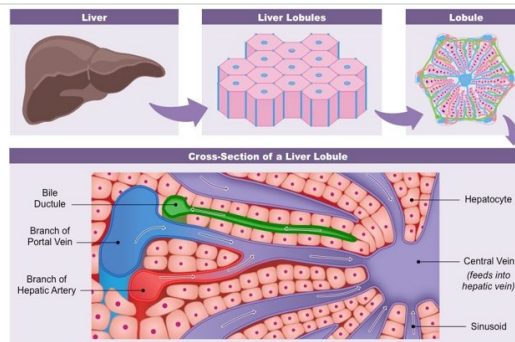
Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Introduction & Blood Supply

Feature	Detail
Size	Largest internal organ — ≈ 1.5 kg in adults
Role	Metabolic hub: processes nutrients, detoxifies substances, produces vital proteins
Portal vein	70–75% of blood flow — nutrient-rich, oxygen-poor, from the GI tract
Hepatic artery	25–30% of blood flow — oxygen-rich, systemic
Drainage	Hepatic veins $\rightarrow$ inferior vena cava

2 Structural & Functional Organization

Lobules	Hexagonal, each with a central vein
Portal triad	Hepatic artery + portal vein + bile duct
Sinusoids	Fenestrated capillaries — exchange between hepatocytes and blood
Kupffer cells	Resident macrophages, immune defense
Bile canaliculi	Channels for bile secretion



Liver lobule structure and the hepatocyte-sinusoid arrangement around the central vein.

3 Metabolism — what the liver does *and* doesn't do

NEVER MISS: exam questions love the "does vs doesn't" trap. The liver **regulates** systemic metabolism but frequently does **not** perform the version of that process you'd expect from other tissues.

Carbohydrate metabolism

Does	Glycogenesis, glycogenolysis, gluconeogenesis; maintains blood glucose homeostasis (70–110 mg/dL)
Does NOT	Use glycogen/glucose primarily for its own energy (unlike muscle) · lacks abundant hexokinase — uses glucokinase instead, so it can't grab glucose at low concentrations the way brain/muscle can · store glycogen for its own use — it's reserved for systemic glucose homeostasis

Lipid metabolism	
Does	Synthesizes cholesterol, triglycerides, lipoproteins; β -oxidizes fatty acids for energy; converts excess carbohydrate/protein into fat
Does NOT	Utilize ketone bodies for energy — lacks thiophorase · store fat long-term (that's adipose tissue) · hydrolyze circulating triglycerides — no lipoprotein lipase

Protein metabolism	
Does	Synthesizes plasma proteins — albumin, clotting factors II, VII, IX, X, XI, V, transport proteins (transferrin, ceruloplasmin, haptoglobin), complement (C1-C9) · urea cycle detoxifies ammonia from amino acid breakdown
Does NOT	Produce immunoglobulins — made by plasma cells, not hepatocytes · synthesize Factor VIII — made by endothelial cells · store protein long-term — muscle does

Clinical correlation: hypoalbuminemia → edema/ascites; coagulopathy in liver failure.

Why the confusion about immunoglobulins? The *fetal* liver does perform hematopoiesis (up to ~6 months gestation), including B-cell precursors — hence some textbooks mention "immune functions of the fetal liver." In the **adult**, the liver makes complement proteins that support antibody function, but never the immunoglobulins themselves.

4 Storage

Glycogen (glucose reserve)

Fat-soluble vitamins A, D, E, K

Vitamin B12 (several years' reserve)

Iron (as ferritin), copper

5 Detoxification

Biotransformation	Phase I (oxidation, reduction) and Phase II (conjugation) reactions process drugs and toxins
Hormone breakdown	Estrogen, cortisol, thyroid hormones
Clinical note	Drug toxicity risk rises in liver disease — e.g. paracetamol overdose

6 Bile Secretion

Composition	Hepatocytes secrete bile acids, cholesterol, phospholipids, and bilirubin
Role	Essential for fat digestion and absorption; bile salts drive micelle formation
Bilirubin metabolism	Hemoglobin breakdown → conjugated in the liver → excreted in bile
Disorders	Jaundice — pre-hepatic, hepatic, or post-hepatic

7 Immune & Endocrine Functions

Kupffer cells remove pathogens and debris

Modulates immune tolerance to gut antigens

Produces IGF-1 under growth hormone stimulation

Converts vitamin D to 25-hydroxycholecalciferol

8 Liver Function Tests

Test	Reflects
ALT, AST	Hepatocellular damage
ALP, GGT	Cholestasis
Albumin	Synthetic function
Bilirubin (total, direct)	Conjugation / excretion

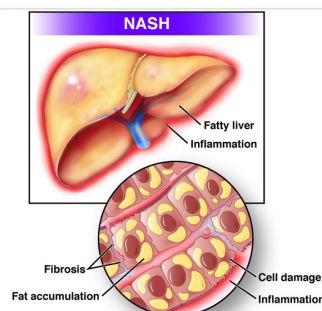
Reading the pattern across these tests is how hepatocellular disease is distinguished from cholestatic disease.

9 Clinical Correlations — quick reference

Cirrhosis	Chronic damage → fibrosis, portal hypertension, ascites
Hepatic encephalopathy	Ammonia accumulation affects brain function
Hepatitis	Viral, autoimmune, or toxic
Hepatocellular carcinoma	Linked to HBV, HCV, cirrhosis
NAFLD	Hepatic manifestation of metabolic syndrome

10 Causes of Liver Damage — the six categories

Category	Examples
Infections	Viral hepatitis A-E; chronic HBV/HCV → cirrhosis/HCC; malaria, leptospirosis, schistosomiasis (endemic areas)
Toxins & drugs	Alcohol (fatty liver → hepatitis → cirrhosis); paracetamol overdose — most common cause of acute liver failure ; anti-TB drugs (isoniazid, rifampin); methotrexate, amiodarone; aflatoxin, <i>Amanita phalloides</i>
Metabolic & genetic	NAFLD; hemochromatosis (iron); Wilson's disease (copper); alpha-1 antitrypsin deficiency; glycogen storage diseases
Immune-mediated	Autoimmune hepatitis; primary biliary cholangitis; primary sclerosing cholangitis (linked with ulcerative colitis)
Vascular	Ischemia ("shock liver"); Budd-Chiari syndrome (hepatic vein thrombosis); right heart failure → passive congestion
Neoplastic	Hepatocellular carcinoma (from hepatitis/cirrhosis); metastases (colon, pancreas, breast, lung)



Fatty liver progressing to inflammation and fibrosis — the metabolic/NAFLD pathway.

P Must-remember pearls

- The liver runs on **glucokinase**, not hexokinase — it can't grab glucose at low blood concentrations the way brain/muscle can, fitting its role as a glucose buffer.
- The liver does **NOT** make immunoglobulins (plasma cells do) or Factor VIII (endothelial cells do) — a classic exam trap.
- **Paracetamol overdose** is the most common cause of acute liver failure.
- ALT/AST → hepatocellular damage; ALP/GGT → cholestasis. That pattern is how LFTs are read.
- Portal vein supplies 70–75% of hepatic blood flow (nutrient-rich, low O₂); hepatic artery supplies the rest (O₂-rich).

Q Test yourself

1. A patient in acute liver failure after a large paracetamol ingestion illustrates that paracetamol overdose is:

- | | |
|---|---|
| A. A rare cause of liver injury | B. The most common cause of acute liver failure |
| C. Only dangerous in patients with pre-existing cirrhosis | D. Metabolized entirely by the kidney |

2. A liver panel showing markedly raised ALP and GGT, with only mildly raised ALT/AST, suggests:

- | | |
|-----------------------------------|----------------|
| A. Hepatocellular damage | B. Cholestasis |
| C. Isolated synthetic dysfunction | D. Hemolysis |

3. A patient with severe liver failure has normal serum immunoglobulin levels because immunoglobulins are produced by:

- | | |
|-----------------|----------------------|
| A. Hepatocytes | B. Kupffer cells |
| C. Plasma cells | D. Endothelial cells |

4. The liver cannot use ketone bodies for its own energy because it lacks which enzyme?

- | | |
|-----------------------|----------------|
| A. Hexokinase | B. Thiophorase |
| C. Lipoprotein lipase | D. Glucokinase |

5. A patient with Budd-Chiari syndrome has thrombosis of which vessel?

- | | |
|-----------------|-----------------------------|
| A. Portal vein | B. Hepatic artery |
| C. Hepatic vein | D. Inferior mesenteric vein |

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1 — B** — paracetamol overdose is the most common cause of acute liver failure.
- 2 — B** — a disproportionate rise in ALP/GGT relative to ALT/AST is the cholestatic pattern.
- 3 — C** — immunoglobulins are made by plasma cells (terminally differentiated B-lymphocytes), not hepatocytes.
- 4 — B** — thiophorase is required to activate ketone bodies for oxidation, and the liver lacks it.
- 5 — C** — Budd-Chiari syndrome is hepatic vein thrombosis, causing outflow obstruction.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam